

ALEXANDER E. SUGIAN

Computer Engineer · Embedded Engineer · Software Engineer

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PROFESSIONAL SUMMARY

- Embedded Systems Engineer and Software Developer with 10+ years of progressive experience in firmware engineering, real-time systems, Linux-based embedded platforms, and full-stack software development across the Philippines and Canada.
- Hands-on track record building safety-critical autonomous systems, including a Smart Disaster Response Drone Network (SDRDN) featuring autonomous mesh networking, AI-powered survivor detection, real-time telemetry, and fail-safe mission command architecture, directly applicable to aerospace onboard computing and high-altitude platform development.
- Proficient in Python and JavaScript (React, Next.js) as primary development languages, with a strong foundation in hardware-software integration, fault-tolerant logic, Linux-based embedded systems, and system bring-up across industrial and commercial deployments.
- Skilled in cybersecurity (CEH, CySA+), OWASP standards, and enterprise IT/OT infrastructure, bringing a rare combination of software engineering depth, systems-level technical breadth, and safety-critical embedded experience, eager to contribute to advanced aerospace software development in near-space environments.

CORE COMPETENCIES / SKILLS

- **Languages (Primary):** Python · JavaScript · C · C++
- **Embedded & Firmware:** C/C++ · Embedded C · Real-Time Systems · Fault-Tolerant Logic · Hardware-Software Interfacing · System Bring-Up · GDB
- **Autonomous & Drone Systems:** ArduPilot/PX4 · ROS/ROS2 · Mesh Networking · Real-Time Telemetry · Fail-Safe Mission Command
- **AI & Machine Learning:** Machine Learning Fundamentals · AI-Powered Detection · Edge Inference · Cisco AI Certified
- **Linux & Low-Level:** Linux Embedded Systems · Low-Level Networking · Performance Profiling · Kernel Interfacing
- **Software & Web Dev:** Django · React · Next.js · Laravel · Vue · PERN Stack · PostgreSQL · REST APIs
- **Testing & DevOps:** Firmware Validation · Automated Testing · Git · CI/CD · Documentation & Config Management
- **Networking & OT:** TCP/IP · DNS · DHCP · VLANs · PLC · HMI · IoT · Sensor Integration
- **Security:** RBAC · OWASP · CEH · CySA+ · Azure AD · VMware

WORK EXPERIENCE

Software Engineer – WFH/Remote

Feb. 2023 to Present

WebRepublik IT Solutions – Bacoar, Cavite Philippines

Key Responsibilities & Achievements:

- Architected and delivered full-stack applications using Django+React, Laravel+Vue, Next.js+Prisma, and PERN stack; primary languages Python and JavaScript throughout all backend and frontend development.
- Built RESTful APIs and microservices integrated with third-party enterprise platforms, improving system throughput and enabling seamless data flow across services.
- Optimized complex SQL queries and database schemas (PostgreSQL/MySQL), reducing query execution time by up to 60%.
- Implemented RBAC, OWASP security standards, and data encryption ensuring regulatory compliance across all applications.
- Automated business workflows using Microsoft Power Platform, saving client teams 10+ hours per week.
- Managed all code via Git, following branching strategies, pull request reviews, and structured documentation practices.

Embedded Systems Engineer – Onsite
Kapex Engineering – Imus, Cavite Philippines

Nov. 2015 to Feb. 2022

Key Responsibilities & Achievements:

- Developed production-grade embedded firmware in C/C++ and Python on Linux-based embedded systems, with deep focus on real-time event handling, fault-tolerant logic, and debugging across hardware-software boundaries, directly applicable to onboard avionics software in safety-critical environments.
- Designed and built a Smart Disaster Response Drone Network (SDRDN) - a safety critical autonomous system featuring autonomous mesh networking, AI-powered survivor detection, real-time telemetry, and fail-safe mission command architecture using C++, Python, ArduPilot/PX4-inspired autonomy, and ROS/ROS2. System was engineered with redundancy and reliability requirements analogous to aerospace platform software.
- Engineered real-time embedded firmware for a Smart Home Automation System: sensor integration, power systems management, low-level networking, and multi-layer hardware-software interfacing across actuator and control layers.
- Developed embedded firmware (C/C++) for Smart Security Systems including biometric/RFID access control, CCTV/NVR integration, and automated systems, resolving complex hardware-software boundary issues in production deployments.
- Utilized Python extensively for firmware testing, tooling, automation scripting, edge-case simulation, and deployment validation across safety-critical embedded environments.
- Maintained all code under Git version control; authored technical documentation covering system architecture, firmware behavior, and deployment procedures for each project.
- Collaborated with hardware and operations teams for system bring-up, performance profiling, and in-field issue resolution across embedded and automation systems.

IT/OT Support Specialist – Onsite

Nov. 2022 to Feb. 2025

GE Vernova (LM Wind Power) – Gaspe, Quebec Canada

Key Responsibilities & Achievements:

- Delivered Tier 1–2 IT/OT support across industrial and corporate environments; managed Windows, macOS, Linux, and mobile endpoints.
- Administered Microsoft 365, Active Directory, Azure AD, VMware, and Citrix environments supporting multi-site operations.
- Managed OT/industrial production network infrastructure including TCP/IP, DNS, DHCP, VLANs, and wireless systems in an industrial context.
- Administered endpoints via Microsoft Intune and Ivanti MDM; executed device imaging, deployment, configuration, and secure decommissioning.
- Tracked incidents, assets, and access permissions through ServiceNow, maintaining structured documentation throughout.

TRAINING AND CERTIFICATES

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|---------------------------------------|---|
| • Google Data Analysis by Python | • IT Support Specialist Professional |
| • Cisco C++ Advance | • Cisco AI Fundamentals Machine Learning |
| • Certified Ethical Hacker (CEH) v.12 | • Cisco Industrial IoT & Control Systems in Manufacturing |

EDUCATION

T.I.P MANILA - Bachelor of Science in Computer Engineering

Completed 2025

Learn Computer Today (LCT) – Electronics Systems Technology

Completed 2021